

Roll No. 

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## Mid Semester Examination March 2025

Name of the Program: B. Tech

Semester: VIII

Name of the Course: Electrical Engineering (Robotics)

Course Code: TOE 805

Time: 90 Minutes Maximum Marks: 50

Note:-

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

|           |                                                                                                                                                                                                                                                                                                |                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Q1</b> | <b>(10marks)</b>                                                                                                                                                                                                                                                                               |                  |
| (a)       | <p>A frame F has been moved nine units along the x-axis and five units along the z-axis of the reference frame. Find the new location of the frame:</p> $F = \begin{bmatrix} 0.527 & -0.574 & 0.628 & 5 \\ 0.369 & 0.819 & 0.439 & 3 \\ -0.766 & 0 & 0.643 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ | <b>CO2</b>       |
|           | <b>OR</b>                                                                                                                                                                                                                                                                                      |                  |
| (b)       | Define different types of robotic configurations for positioning the hand of a robot.                                                                                                                                                                                                          |                  |
| <b>Q2</b> | <b>(10marks)</b>                                                                                                                                                                                                                                                                               |                  |
| (a)       | Explain various characteristics of robots.                                                                                                                                                                                                                                                     |                  |
|           | <b>OR</b>                                                                                                                                                                                                                                                                                      |                  |
| (b)       | A point P (2,3,4) <sup>T</sup> is attached to a rotating frame. The frame rotates 90° about the x-axis of the reference frame. Find the coordinates of the point relative to the reference frame after the rotation and verify the result graphically.                                         | <b>CO1&amp;2</b> |
| <b>Q3</b> | <b>(10marks)</b>                                                                                                                                                                                                                                                                               |                  |
| (a)       | How can you define a robot? How can we classify the robots according to Japanese Industrial Robot Association?                                                                                                                                                                                 | <b>CO1</b>       |
|           | <b>OR</b>                                                                                                                                                                                                                                                                                      |                  |
| (b)       | State various applications of robots.                                                                                                                                                                                                                                                          |                  |
| <b>Q4</b> | <b>(10marks)</b>                                                                                                                                                                                                                                                                               | <b>CO1</b>       |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                    |           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| (a) | State advantages and disadvantages of Robots(five each).                                                                                                                                                                                                                                                                                                                                                           |           |
|     | OR                                                                                                                                                                                                                                                                                                                                                                                                                 |           |
| (b) | Explain the basic components involved to form a robot.                                                                                                                                                                                                                                                                                                                                                             |           |
| Q5  | (10 Marks)                                                                                                                                                                                                                                                                                                                                                                                                         | CO1<br>&2 |
| (a) | <p>A point P (7,3,2)<sup>T</sup> is attached to a frame ( <math>\bar{n}, \bar{o}, \bar{a}</math> ) and is subjected to the below mentioned transformations. Find the coordinates of the point relative to the reference frame after the transformations.</p> <p>(1) Rotation of 90° about z-axis</p> <p>(2) Followed by a rotation of 90° about the y axis</p> <p>(3) Followed by a translation of [ 3, 3, 3 ]</p> |           |
|     | OR                                                                                                                                                                                                                                                                                                                                                                                                                 |           |
| (b) | What are the different types of joints in robots? Discuss in detail.                                                                                                                                                                                                                                                                                                                                               |           |